

AI-Powered Financial Fraud Detection System

MBA AI Project (Project 10) — Machine Learning model to detect fraudulent credit card transactions.

Overview

Yeh project credit card transactions ke data ko analyze karke fraud transactions ko normal transactions se alag karta hai, Machine Learning classification models ka use karke.

Tech Stack

- Python
- Google Colab
- pandas, numpy

- scikit-learn
- matplotlib, seaborn



Dataset

[Kaggle Credit Card Fraud Detection Dataset](#)

- 284,807 transactions, jisme se sirf 492 fraud hain (~0.17%)
- Features `V1` to `V28` (PCA transformed for privacy), `Time`, `Amount`, aur `Class` (target: 0 = normal, 1 = fraud)



How to Run

1. `Fraud_Detection_Project.ipynb` ko Google Colab mein open karo
2. Apni Kaggle API key (`kaggle.json`) upload karo jab pucha jaaye (ya file manually upload karo)
3. Sabhi cells ko top se bottom tak run karo



Models Used

1. **Logistic Regression** — baseline model
2. **Random Forest Classifier** — improved performance, class imbalance handle karne ke liye

```
class_weight='balanced'
```



Evaluation Metrics

- Accuracy, Precision, Recall, F1-Score, ROC-AUC
- Recall par extra focus, kyunki fraud detection mein missed fraud (false negatives) sabse costly hote hain



Project Structure

```
|— Fraud_Detection_Project.ipynb
# Main notebook (EDA + Model
  Training + Evaluation)
|— README.md
# Project documentation
```

```
└─ requirements.txt  
# Python dependencies
```

Future Scope

- SMOTE / other resampling techniques for imbalance
- XGBoost / LightGBM models
- Deployment as a Flask/Streamlit web app for real-time fraud checking

Author

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